One Trellis, Two Assessments: An Embedded CTC–Viterbi Alternative to MFA for Phoneme and Lexical-Stress Scoring

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Abstract—Pronunciation-assessment systems often run an external forced aligner before computing phoneme goodness and syllable-stress features. We investigate whether one resident phoneme Connectionist Temporal Classification (CTC) model can do both jobs. Our system decodes the maximum-probability path through a canonical phone-conditioned CTC trellis, uses phone-state frames for mean log-posterior scores, and converts the same path into continuous regions for a fixed Mallela-inspired sequential stress scorer. Montreal Forced Aligner (MFA) 3.4.2 is the external-alignment baseline. In a controlled SpeechOcean762 experiment, the CTC and MFA conditions share the acoustic posteriors, targets, score, calibration, and stress model; only phone-region estimation differs. On 43,613 paired test phones, CTC–Viterbi obtains calibrated Pearson correlation 0.459 versus 0.317 for MFA. The paired difference is 0.142 (95% speaker-bootstrap confidence interval 0.120–0.164). On 1,575 polysyllabic words whose human stress grade is correct, canonical stress-location accuracy is 74.9% versus 69.5%, a 5.4-point difference (95% interval 2.8–8.0). A complete paired batch—one CTC encoding, 43,613 phone scores, and 1,621 polysyllabic stress predictions per condition—takes 98.6 ms per utterance with CTC–Viterbi and 128.8 ms with a one-job MFA baseline, a 1.31× speed ratio. The results support an MFA-free architecture without an observed accuracy trade-off under this protocol.

Index Terms—pronunciation assessment, goodness of pronunciation, CTC, Viterbi decoding, lexical stress, forced alignment

I. Introduction

Computer-assisted pronunciation training benefits from both segmental feedback—which phone was weak—and suprasegmental feedback—which syllable received lexical stress. Classical goodness-of-pronunciation (GOP) systems first align a known transcript and then aggregate acoustic-model evidence inside each phone interval [1], [2]. Stress systems likewise require temporal regions because duration, intensity, pitch, and sonority are properties of a syllable or its nucleus rather than a complete utterance [3], [4].

Montreal Forced Aligner (MFA) provides reliable, general-purpose word and phone intervals through a Kaldi-based alignment pipeline [5], [6]. It also introduces a second acoustic model, a pronunciation dictionary, a database, subprocess orchestration, and a handoff of alignment artifacts. Those costs are reasonable for corpus preparation but unattractive for an interactive tutor already running a phoneme CTC model.

This work asks whether the resident CTC model can supply the necessary phone regions itself. CTC was designed to train sequence models without frame-level labels by marginalizing paths that collapse to an output sequence [7]. At inference, the same topology supports a maximum-probability path constrained by the expected phone sequence. We call this operation embedded CTC–Viterbi alignment. It is not unconstrained phone recognition: the canonical phones define the trellis. The architectural claim is narrower and operational: no external MFA/Kaldi alignment job is used by an assessment request.

Figure 1 contrasts the two pipelines. Our contributions are:

- a tested CTC–Viterbi decoder whose phone regions drive both phoneme scoring and lexical-stress features;
- a paired experiment that holds all downstream scoring components fixed while replacing only CTC regions with MFA regions;
- a matched complete-pipeline latency experiment that charges each condition for phone scoring and lexical-stress inference; and
- checked-in phone- and word-level outputs, failure records, environment metadata, and speaker-bootstrap inference.

II. Related Work

Witt and Young introduced phone-level pronunciation confidence for language learning [1]; later DNN-GOP formulations replaced generative likelihoods with neural posteriors [2]. SpeechOcean762's released Kaldi recipe constructs an alignment graph directly from the expert-selected canonical phone sequence and fits phone-specific regressors [8]. Cao et al. show that CTC also supports alignment-marginalized scalar and vector GOP definitions [9]. Our objective differs: we deliberately recover a single CTC path because both phone scoring and stress feature extraction require localized evidence, while eliminating a separate aligner service.

Lexical stress is conveyed by relative duration, intensity, and fundamental frequency [3]. Yarra et al. demonstrate that automatic boundary quality affects stress detection and examine forced-alignment acoustic models and lexicons on ISLE [4]. Mallela et al. treat a word as a sequence

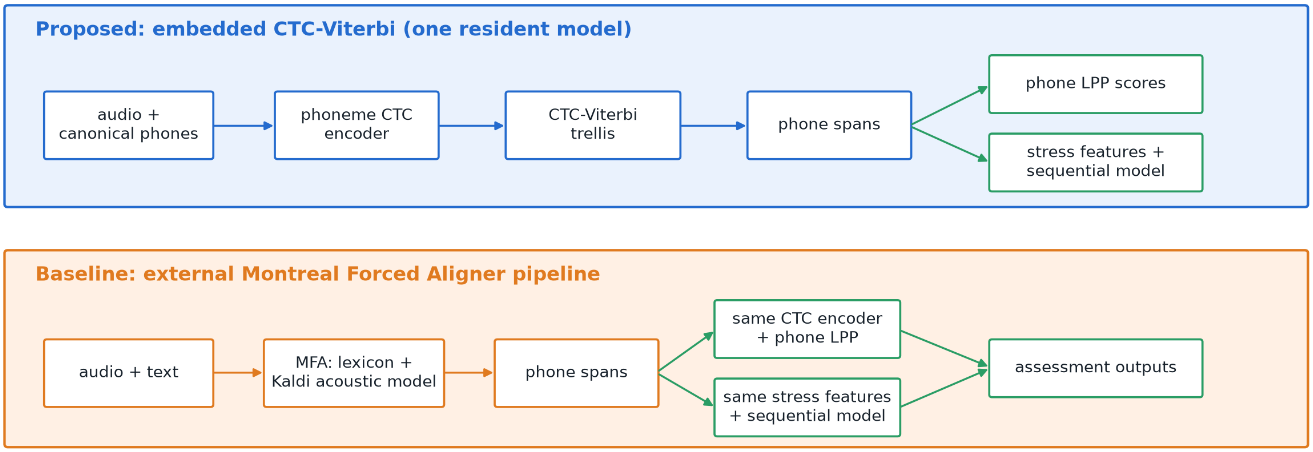

Fig. 1. The proposed path obtains phone scores and stress regions from one resident CTC model. The baseline first runs external MFA, then passes its intervals to the same CTC score and stress modules.

of syllable features, use masking for variable-length words, and compare RNN, LSTM, GRU, and attention models on German and Italian learners in ISLE [10]. Their pipeline obtains phone alignments automatically, manually corrects them, derives syllables, and applies a word-level post-processing rule that selects one primary stress. We reuse the sequential and word-constraint ideas, but our local checkpoint and feature code are an adaptation rather than a reproduction of their ISLE model.

III. Method

A. Embedded CTC–Viterbi path

Let $\mathbf{L} = (l_1, \ldots, l_U)$ be the expected phone sequence and let $\ell_{t,k} = \log p(k \mid \mathbf{x}, t)$ be the log posterior for token k at CTC frame t. We form the expanded state sequence

$$\mathbf{z} = (\phi, l_1, \phi, l_2, \ldots, \phi, l_U, \phi), \tag{1}$$

where ϕ is blank. The Viterbi recurrence is

$$V_t(s) = \ell_{t,z_s} + \max_{r \in \mathcal{P}(s)} V_{t-1}(r), \tag{2}$$

where $\mathcal{P}(s)$ permits staying at s, advancing from $s-1$, and advancing from $s-2$ into a phone state only when the skipped and current phone labels differ. The last restriction forces repeated adjacent phones to pass through blank. Initialization permits the leading blank or first phone; termination permits the last phone or trailing blank. Backtracking yields one state per frame in $\mathcal{O}(TU)$ time and $\mathcal{O}(TU)$ memory.

The implementation is verified against exhaustive enumeration of every valid path in small lattices. It also includes a dedicated repeated-phone test.

B. Phone regions and score

Let $F_i = \{t : z_{s_t} = l_i\}$ be the frames assigned to phone i. Its scalar score is the frame-averaged log posterior

$$g_i = \frac{1}{|F_i|} \sum_{t \in F_i} \ell_{t,l_i}. \tag{3}$$

For a continuous interval, the start and end of the phone-state run are extended across adjacent blank gaps by midpoint splitting. Equation 3 uses only phone-state frames, whereas the expanded intervals are passed to the stress branch.

For the MFA baseline, MFA supplies phone intervals. Frame centers falling inside an MFA interval replace F_i in Eq. 3; a nearest-frame rule handles an interval shorter than one CTC frame. A global edit-distance match pairs identical phones between the expert canonical sequence and the MFA output. Only phones matched in both conditions enter the comparison.

C. Stress scoring

Canonical pronunciations are syllabified by the maximal-onset principle. Phone regions are grouped by syllable and the vowel-phone interval defines the nucleus. The local feature extractor emits 19 acoustic measurements covering duration, pitch, intensity, and spectral properties and 19 context features. The fixed scorer contains three LSTM layers (64, 32, and 16 units), a four-unit attention projection, and a sigmoid output. Its standardized neural scores are combined with a relative acoustic-prominence score; word-level argmax selects exactly one primary-stress location. This same scorer, scaler, audio, pronunciation, and post-processing are used for CTC and MFA intervals.

IV. Experimental Design

A. Corpus and models

SpeechOcean762 contains 5,000 English utterances from 250 Mandarin-speaking learners, divided into speaker-disjoint sets of 2,500 utterances from 125 speakers each [8]. Five experts supply phone accuracy scores from 0 to 2 and word stress scores: 10 for correct stress (or a monosyllable) and 5 for incorrect stress. We retain the official split.

The resident model is mostafaashahin/wav2vec2-base-timit-phoneme-arpa-39, a Wav2Vec 2.0 base CTC phone model [11]. MFA 3.4.2 uses its english_us_arpa dictionary and acoustic model. With –ignore_oovs, MFA exports 2,462 training and 2,447 test TextGrids. The paired phone sets contain 43,594 training and 43,613 test phones. The CTC system itself scores all 2,500 test utterances; restriction is required only for the paired baseline comparison.

B. Phone endpoint

For each condition, phone-specific quadratic regressors map Eq. 3 to the human grade. Regressors are fitted only on the matched official training phones; a global quadratic is used if a phone has fewer than 20 examples. Predictions are clipped to $[0, 2]$. We report Pearson correlation (PCC) and Spearman correlation (SRCC) on the paired test phones. To preserve dependence within a speaker, the central PCC difference is evaluated with 2,000 paired speaker bootstrap samples.

C. Stress endpoints

The stress comparison includes polysyllabic words whose complete phone sequence is matched to MFA, yielding 1,621 words: 1,575 rated correct and 46 rated as stress errors. The primary boundary-sensitivity endpoint is canonical stress-location accuracy among human-correct words; here the expert label establishes that the realized stress should agree with the dictionary location. We bootstrap its paired CTC-minus-MFA difference by speaker 2,000 times. As a secondary endpoint, the predicted probability assigned to the canonical primary syllable is used to distinguish human-correct from human-incorrect stress, reported as area under the ROC curve (AUROC). SpeechOcean does not annotate the realized destination of a misplaced stress, so wrong-location accuracy cannot be measured for the 46 errors.

D. Latency protocols

All timings use the same ARM64 macOS CPU and a complete paired batch workload. For each of the 2,447 MFA-aligned test utterances, both conditions execute one resident CTC encoding, score the same matched phone tokens, and run the same stress model on every fully matched polysyllabic word. The proposed condition additionally runs the CTC–Viterbi trellis; the baseline uses an MFA TextGrid. Condition order alternates by utterance after one excluded warm-up to reduce systematic order and thermal effects. Each condition uses one top-level Python assessment process and the same four PyTorch intra-operation CPU threads.

The baseline is also charged for a fresh MFA 3.4.2 corpus alignment with –num_jobs 1. This stage processes all 2,500 inputs, including the 53 cases for which no TextGrid is exported, and takes 89.3 s. Complete serial wall time is the measured MFA alignment time plus measured downstream assessment time. We divide each condition's total by the 2,447 paired outputs. This design tests the deployed assessment workload rather than alignment-only throughput.

TABLE I
Paired phone assessment on 43,613 held-out phones.

Metric	CTC–Viterbi	MFA
Raw PCC	0.401	0.216
Raw SRCC	0.325	0.233
Quadratic PCC	0.459	0.317
Quadratic SRCC	0.376	0.302

TABLE II
Paired stress results on 1,621 polysyllabic words.

Metric	CTC–Viterbi	MFA
Canonical location accuracy†	74.9%	69.5%
Correct-stress AUROC	0.726	0.695
Grade PCC	0.155	0.133
Grade SRCC	0.144	0.122

†Among 1,575 words rated correct by humans.

V. Results

Table I shows higher phone correlation for CTC–Viterbi. The paired PCC difference is 0.142, with a 95% speaker-bootstrap interval of [0.120, 0.164]. Since the interval lies above zero, the result establishes no observed accuracy trade-off and favors the embedded regions under this protocol. CTC and MFA boundaries differ by 50.0 ms median; mean disagreement is 150.7 ms because a minority of alignments have large offsets.

For canonical stress location (Table II), the paired difference is 5.4 percentage points and the 95% speaker-bootstrap interval is [2.8, 8.0] points. Correct-stress discrimination also favors CTC descriptively. Because only 46 paired polysyllabic words carry the error grade, AUROC and grade correlations should be treated as secondary, high-variance estimates.

Table III shows that the embedded path is 1.31× faster for the complete matched workload. CTC–Viterbi region construction costs more than reading and indexing existing MFA intervals, but it replaces the 89.3-s external alignment stage. The two encodings are nearly identical in cost, as expected, and the same stress workload is executed in both conditions. An exploratory four-job MFA run reached 28.0 ms per input for alignment alone; because it excludes the shared CTC phone scorer and stress branch, it is not used as the end-to-end comparison.

TABLE III
Mean complete-pipeline latency per paired utterance. MFA alignment is the one-job corpus wall time amortized over 2,447 outputs.

Component	CTC–Viterbi	MFA	MFA/CTC
CTC encoding	62.9 ms	62.8 ms	—
Phone regions + LPP	4.4 ms	0.2 ms	—
Stress branch	31.3 ms	29.4 ms	—
External alignment	—	36.5 ms	—
Complete pipeline	98.6 ms	128.8 ms	1.31×

The speed claim remains specific to this hardware, worker configuration, and corpus.

VI. Discussion and Limitations

The result suggests that a CTC model trained to recognize phones can localize the evidence most useful to its own posterior score better than cross-system MFA intervals. This does not imply that its timestamps are closer to manual articulatory boundaries. MFA is the reference pipeline, not ground truth, and the median 50-ms disagreement cannot decide which boundary is physically correct. The experiment instead asks an application-level question: which interval source yields better human-correlated assessments when downstream components are held fixed?

The paired design removes many confounds but has limitations. MFA OOV failures exclude 38 training and 53 test utterances. Global phone-sequence matching removes unmatched tokens from both conditions, potentially favoring easier examples. The CTC acoustic model was trained for TIMIT-style phone recognition, not specifically for L2 assessment. The quadratic calibration may not transfer to another learner population. An external-aligner baseline using a native phone model matched more closely to the CTC inventory could perform differently.

The stress experiment validates a boundary substitution in the repository's fixed Mallela-inspired scorer; it does not reproduce Mallela et al.'s manually corrected ISLE experiment. SpeechOcean supplies only correct/incorrect word stress, not the realized wrong stress location. A future study should repeat the paired comparison on ISLE with speaker-disjoint German/Italian folds, manual syllable boundaries, and explicit wrong-location labels. Finally, a parallel MFA deployment and a batched or parallel CTC service could change throughput; worker-count scaling should be measured separately rather than mixed into the one-job method comparison.

Terminology also deserves care. Some authors use forced alignment for any transcript-conditioned best path, including Eq. 2. Our engineering distinction is between an embedded path in the assessment model and the external MFA/Kaldi pipeline. Accordingly, the strongest portable terms for the proposal are MFA-free and external-aligner-free.

VII. Conclusion

We replaced peak timing and the external alignment dependency with one canonical-conditioned CTC–Viterbi path shared by phoneme GOP and lexical-stress assessment. On paired SpeechOcean762 data, the embedded path improves both phone correlation and canonical stress-location accuracy relative to MFA regions, with speaker-bootstrap intervals excluding an accuracy loss. It also reduces complete one-job phone-and-stress processing time from 128.8 to 98.6 ms per paired utterance. The proposed architecture is therefore a faster single-model path under the measured protocol, not a universal claim about all hardware, worker counts, acoustic models, or corpora.

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